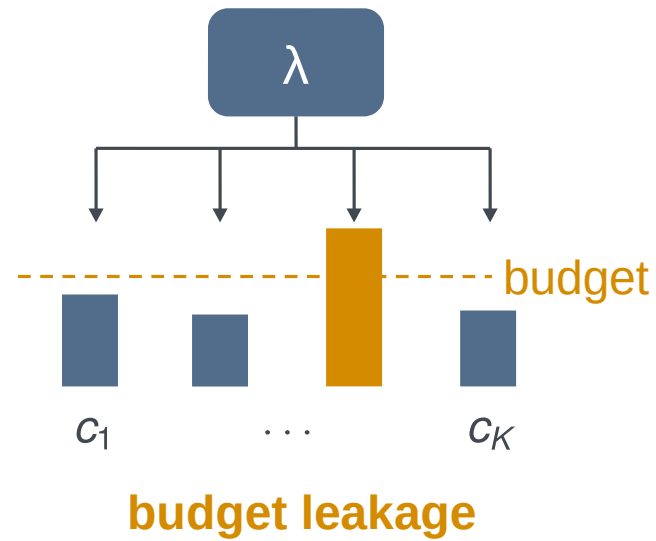
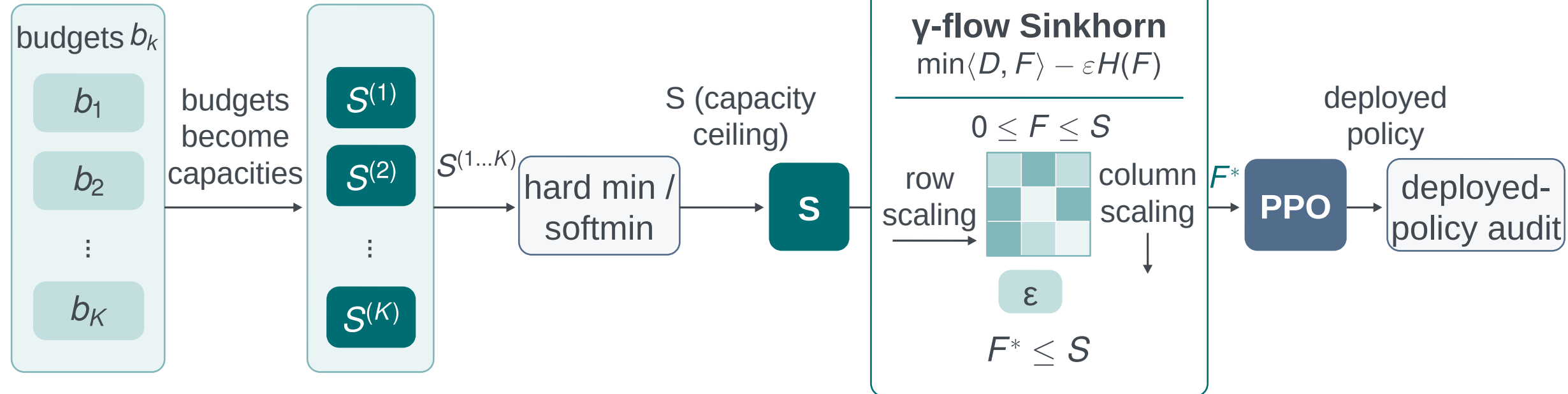


Penalty-based safe RL

$$\max_{\pi} J_r(\pi) - \lambda^T J_c(\pi)$$

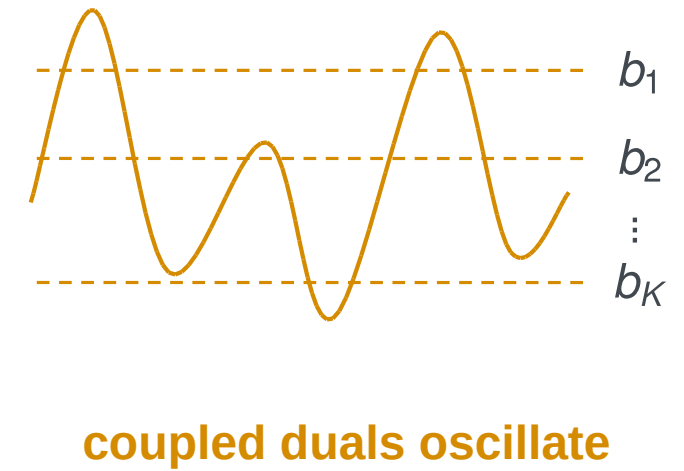


SafeTransport



Constraint-based safe RL

$$\max_{\pi} J_r(\pi) \text{ s.t. } J_{c_k}(\pi) \leq b_k, k = 1 \dots K$$



penalty-like

PPO / SAC

unconstrained or penalty

PPO-Lag / FOCOPS

external safety controller

SafeTransport

one inner capacity projection

CPO

local constrained step

CUP / hard projection

constraint-dominant

constraint-like